

BlockBase – Day 3

Mentored by FEC

Deploying, Running & Interacting

The Deploy & Run module is for sending transactions to the current Environment.

The three main actions of the Deploy & Run module are:

1. Deploying a contract
2. Accessing an on chain contracts
3. Interacting with the functions of a deployed contract

Today's task will cover all of these.

To deploy a contract, you need to have a contract compiled. To check that there is a compiled contract, look in the CONTRACT select box under the VALUE input field.

If nothing is there, you'll need to compile a contract. If you do not see your desired contract in the CONTRACT select box list, make sure the file with the contract is the active tab in the Editor.

DEPLOY & RUN TRANSACTIONS ✓ >

ENVIRONMENT 

Remix VM (Cancun)  

VM

ACCOUNT 

0x5B3...eddC4 (100 ether)   

GAS LIMIT

☒ Estimated Gas

☐ Custom

3000000

VALUE

0

Wei 

CONTRACT

Owner - contracts/2_Owner.sol 

evm version: cancun

Deploy

☐ Publish to IPFS

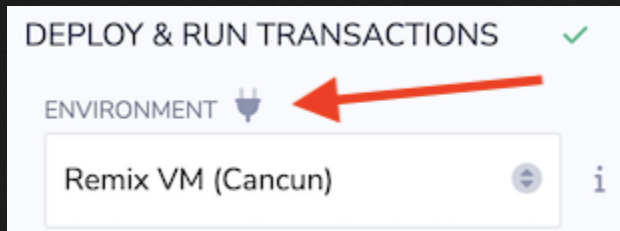
At Address

Load contract from Address

Once the contract is selected, choose the chain for deployment and/or method for deployment in the ENVIRONMENT select box.

NOTE: If you want to connect Remix with a browser wallet (like Metamask) you should use the Injected Provider. Then in the browser wallet, you can select your desired chain when you have input its configuration.

For assistance with getting the correct configuration (the RPC server address, etc.), click the plug icon next to the word ENVIRONMENT to go to chainlist.org. (Choose this to be WalletConnect, when using MetaMask)



Gas Limit:

- This sets the maximum amount of gas allowed to be spent for all the contract transactions created in Remix. (We will learn in depth about gas on day 4)

Value:

- This sets the amount of ETH, WEI, GWEI, etc. that is sent to a contract or a payable function. (these are primarily used for transactions in the smart contract)

Note: payable functions have a red button.

The Value field is always reset to 0 after each transaction execution.

The Value field is NOT for gas.

DEPLOY AND RUN TRANSACTIONS

Environment ⓘ

Account ⓘ ⓘ ⓘ

Gas limit

Value ⓘ ⓘ

ⓘ

▼

or

Deploy and AtAddress

- In the image above, the select box is set to Ballot. This select box will contain the list of compiled contracts.
- **Deploy** sends a transaction that deploys the selected contract. When the transaction is mined, the newly created instance will be added (this might take several seconds).
Note: If the contract's constructor function has parameters, you will need to specify them.
- **AtAddress** is used to access a contract that has already been deployed. Because the contract is already deployed, accessing a contract with AtAddress does not cost gas.

Note: When using AtAddress, be sure you trust the contract at that address.

To use AtAddress, you need to have the source code or ABI of the deployed contract in the active tab of the Editor. When using the source code, it must be compiled with the same compilation settings as the deployed contract that you are trying to access.

Using the ABI with AtAddress

The ABI is a JSON array which describes the contract's interface.

To interact with a contract using the ABI, create a new file in Remix with extension *.abi and copy the ABI content into it.

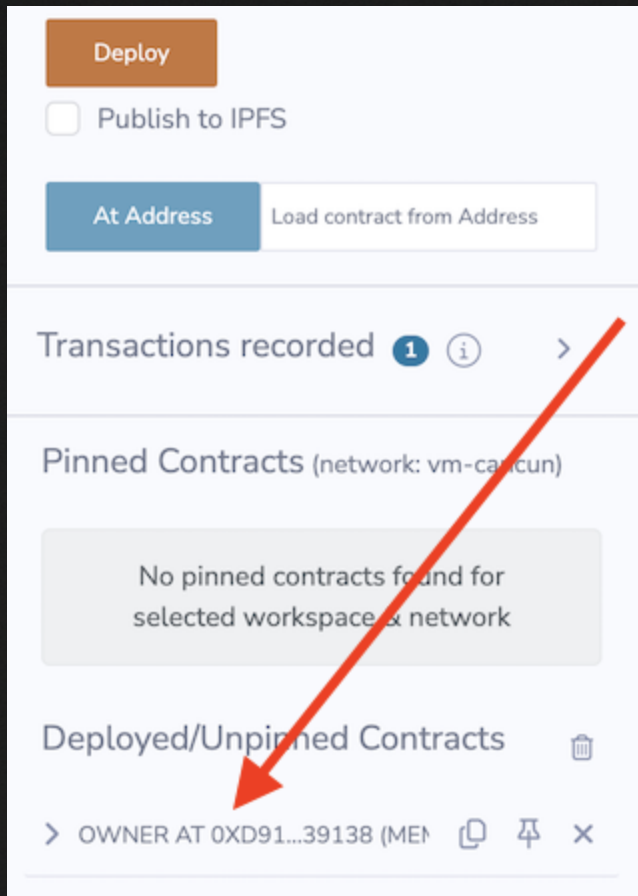
Make sure this file is the active tab in the Editor. Then, in the field next to **At Address**, input the contract's address and click on **At Address**. If successful, an instance of the contract will appear below in the list of Deployed Contracts.

Note: To generate the ABI, in the Solidity compiler after a contract is compiled, click on the Compilation Details button. A modal will come up that contains the ABI, among other info.

Interacting with deployed contracts

After a contract is deployed or after it is accessed with AtAddress, the deployed instance will appear in the Deploy/Unpinned Contracts section at the bottom of the Deploy & Run panel.

The deployed contract's address is visible as are a few other icons – one of which is the pin icon.



Pinned contracts

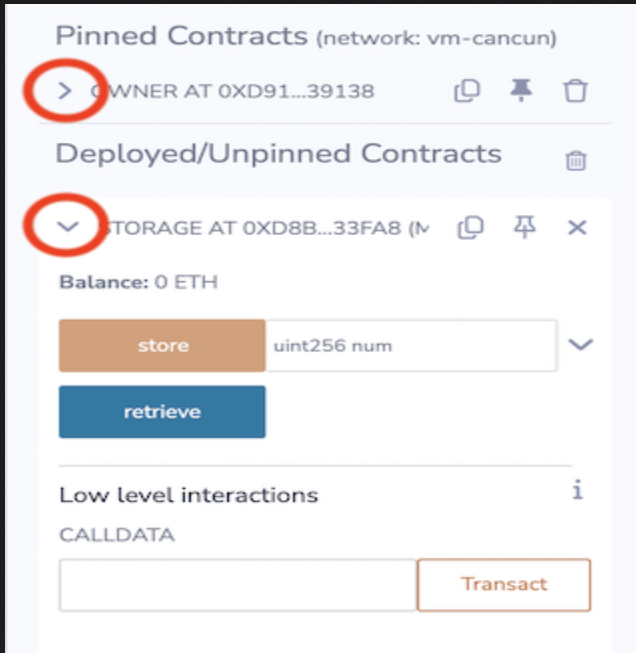
When a contract is pinned, it will jump up to the Pinned Contracts section and Remix will save the contract's address and the ABI (in the .deploys folder of the current Workspace). When Remix is refreshed, the pinned contracts will be loaded into the Pinned Contracts section.

Pinned contracts are chain & Workspace specific

Because a pinned contract's address and ABI are stored in a File Explorer Workspace, the same Workspace must be active to see its pinned contracts. Similarly, only the pinned contracts of the currently selected chain will show.

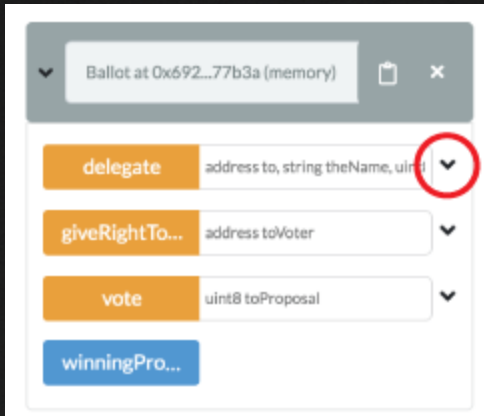
Functions

To see a contract's functions, click the caret on the left side of the panel.



The functions' buttons can have different colors.

- Blue buttons are for **view** or **pure** functions. Clicking a blue button does not create a new transaction – so there will be no gas fees.
- Orange buttons are for **non-payable** functions. Non-payable functions change the state of the contract BUT do not accept value (typically ETH) being sent with the transaction. Clicking an orange button will create a transaction and will cost gas.
- Red buttons are for **payable** functions. Clicking a red button will create a new transaction and this transaction can accept a value (typically ETH). The amount of value is set in the Value field which is under the Gas Limit field.



Inputting parameters in the collapsed view

In the collapsed view:

Parameters are separated by commas.

In the example above the “delegate” function has 3 parameters.

Inputting parameters in the expanded view

Clicking the ‘down’ caret brings you to the expanded view – where parameters are input one at a time.

